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(54) **PYRIDINE DERIVATIVES SUBSTITUTED BY HETEROCYCLIC RING AND AMINO GROUP**

DURCH EINEN HETEROCYCLISCHEN RING UND EINE AMINOGRUPPE SUBSTITUIERTE
PYRIDINDERIVATE

DÉRIVÉS DE PYRIDINE SUBSTITUÉS PAR UN NOYAU HÉTÉROCYCLIQUE ET UN GROUPE
AMINO

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US-A1- 2003 045 554 US-A1- 2007 105 904
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- **DATABASE Pubmed U.S. National Library of Medicine; 30 November 2012 (2012-11-30), "3-[3-[[4-(Pyridin-2-yloxy)methyl]phenyl]methyl]-1,2-oxazol-5-yl]pyridin-2-yl]carbamic acid", Database accession no. 68452548**

Remarks:

The file contains technical information submitted after the application was filed and not included in this specification

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Description

BACKGROUND

[0001] Management of opportunistic infections has remained an ongoing challenge due to chemotherapy-induced immunodeficiency and an increasing elderly population.

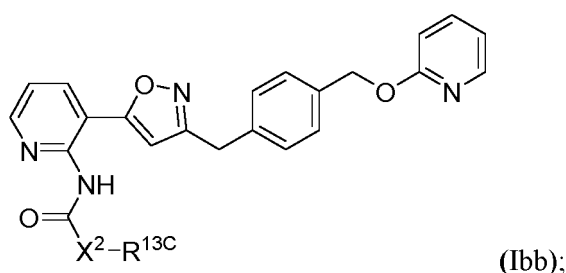
[0002] EP-2351752, EP-1944303 and US-2009/082403 disclose 3-heteroarylpyridines which are said to be antifungal agents suitable for treating humans.

[0003] WO-2015/002080 and WO-2011/051198 disclose 3-heteroarylpyridines which are said to be antifungal agents useful as agrochemicals.

[0004] US-2011/263845 discloses a method for preparing 3-heteroarylpyridines.

BRIEF SUMMARY OF THE INVENTION

[0005] In an aspect, provided herein is a compound of structural Formula (Ibb), or a pharmaceutically acceptable salt, solvate, tautomer, or stereoisomer thereof:



wherein:

X² is -O-, -S-, or -NR^{13B};

R^{13B} and R^{13C} are independently hydrogen, halogen, -CF₃, -CCl₃, -CBr₃, -Cl₃, -COOH, -CONH₂, substituted or unsubstituted alkyl, substituted or unsubstituted heteroalkyl, substituted or unsubstituted cycloalkyl, substituted or unsubstituted heterocycloalkyl, substituted or unsubstituted aryl, or substituted or unsubstituted heteroaryl; or R^{13B} and R^{13C} substituents bonded to the same nitrogen atom may optionally be joined to form a substituted or unsubstituted heterocycloalkyl or substituted or unsubstituted heteroaryl.

[0006] In another aspect, provided herein is a pharmaceutical composition, including a compound as described herein having structural Formula (Ibb), or a pharmaceutically acceptable salt, solvate, tautomer, or stereoisomer thereof, and at least one pharmaceutically acceptable carrier.

[0007] In another aspect, provided herein is a compound for use in a method of treating a fungal disease or infection, the method comprising administering to a subject in need thereof a therapeutically effective amount of a compound as described herein having structural Formula (Ibb), or a pharmaceutically acceptable salt, solvate, tautomer, or stereoisomer thereof. In some embodiments, the fungal disease or infection is caused by a *Cryptococcus*, *Aspergillus*, *Candida*, *Fusarium*, or *Scedosporium* fungus or a fungus from the Mucorales order. In some embodiments, the fungal disease or infection is azole-resistant and/or echinocandin-resistant.

[0008] In some embodiments, the method further comprises administering at least one antifungal agent in combination with the compound of Formula (Ibb), or a pharmaceutically acceptable salt, solvate, tautomer, or stereoisomer thereof. In some embodiments, the at least one antifungal agent is an azole, an echinocandin, amphotericin B deoxycholate, amphotericin B cochleate, 5-fluorocytosine, terbinafine, griseofulvin, VL-2397, ibrexafungerp, orotomide F901318, or combinations thereof. In some embodiments, the azole is ketoconazole, fluconazole, posaconazole, itraconazole, voriconazole, isavuconazole, or miconazole. In some embodiments, the echinocandin is caspofungin, anidulafungin, micafungin, or rezafungin.

[0009] In some embodiments, the compound of Formula (Ibb), or a pharmaceutically acceptable salt, solvate, tautomer, or stereoisomer thereof, and the antifungal agent are administered simultaneously, approximately simultaneously, or sequentially, in any order. In some embodiments, the compound of Formula (Ibb), or a pharmaceutically acceptable salt, solvate, tautomer, or stereoisomer thereof, and the antifungal agent are administered simultaneously or approximately simultaneously. In some embodiments, the compound of Formula (Ibb), or a pharmaceutically acceptable salt, solvate, tautomer, or stereoisomer thereof, and the antifungal agent are administered sequentially. In some embodiments, the

known by those skilled in the art.

[0071] A discussion of prodrugs is provided in Higuchi, T., et al., "Pro-drugs as Novel Delivery Systems," A.C.S. Symposium Series, Vol. 14, and in Bioreversible Carriers in Drug Design, ed. Edward B. Roche, American Pharmaceutical Association and Pergamon Press, 1987.

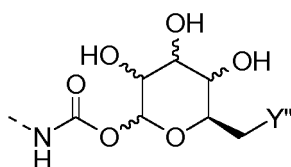
[0072] The term "prodrug" is also meant to include any covalently bonded carriers, which release the active compound *in vivo* when such prodrug is administered to a mammalian subject. Prodrugs of an active compound, as described herein, are prepared by modifying functional groups present in the active compound in such a way that the modifications are cleaved, either in routine manipulation or *in vivo*, to the parent active compound. Prodrugs include compounds wherein a hydroxy, amino or mercapto group is bonded to any group that, when the prodrug of the active compound is administered to a mammalian subject, cleaves to form a free hydroxy, free amino or free mercapto group, respectively. In some embodiments, prodrugs include any group bound to a heteroatom, such as the nitrogen of a pyridine which is cleaved *in vivo* to form the active compound or metabolite thereof. Examples of prodrugs include, but are not limited to, acetate, formate phosphate, and benzoate derivatives of alcohol or amine functional groups in the active compounds and the like.

[0073] Prodrugs undergo some form of a chemical transformation to produce the compound that is biologically active or is a precursor of the biologically active compound. In some cases, the prodrug is biologically active, more or less than the intended active drug itself, and serves to improve drug efficacy or safety through improved oral bioavailability, and/or pharmacodynamic half-life, etc. Prodrug forms of compounds are utilized, for example, to improve bioavailability, improve subject acceptability such as by masking or reducing unpleasant characteristics such as bitter taste or gastrointestinal irritability, alter solubility such as for intravenous use, provide for prolonged or sustained release or delivery, improve ease of formulation, or provide site-specific delivery of the compound. Prodrugs are described in The Organic Chemistry of Drug Design and Drug Action, by Richard B. Silverman, Academic Press, San Diego, 1992, Chapter 8: "Prodrugs and Drug delivery Systems" pp. 352-401; Design of Prodrugs, edited by H. Bundgaard, Elsevier Science, Amsterdam, 1985; Design of Biopharmaceutical Properties through Prodrugs and Analogs, Ed. by E. B. Roche, American Pharmaceutical Association, Washington, 1977; and Drug Delivery Systems, ed. by R. L. Juliano, Oxford Univ. Press, Oxford, 1980.

[0074] Some prodrugs comprise phosphorus moieties including phosphates or derivatives thereof. One such class of prodrugs is the aryl amidate (McGuigan) type. One report disclosed pharmacokinetic evaluation in the cynomolgus monkey of an aryl amidate prodrug of abacavir. (C. McGuigan et al., "Application of phosphoramidate pronucleotide technology to abacavir leads to a significant enhancement of antiviral potency," J. Med. Chem. 2005, 48, 3504-3515).

[0075] The compounds of the present application comprise a prodrug moiety that is carbamate, thiocarbamate or urea to mask an amino group on the active compound. The prodrug moiety of carbamate, thiocarbamate or urea is metabolized *in vivo* to afford the free amino moiety on the active compound.

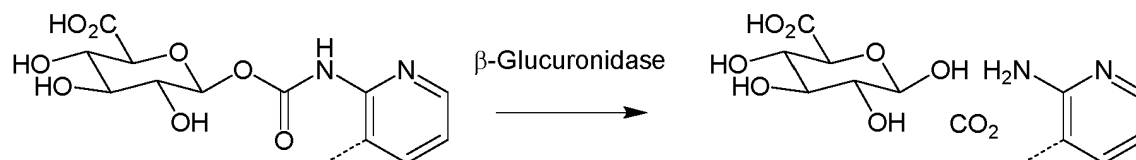
[0076] In some embodiments, glycosyl carbamates (as shown in Formula (S1)) are used as prodrugs to mask an amino group in the active compound.



Formula (S1),

wherein Y'' is OH or CO₂H. The carbohydrate portion of the glycosyl carbamates can be a natural or unnatural hexose or its derivatives. In some embodiments, the hexose is selected from the group consisting of allose, altrose, glucose, mannose, gulose, idose, galactose, and talose. In some embodiments, the hexose has D- configuration. In other embodiments, the hexose has L- configuration. In some embodiments, the hexose has D- configuration. In other embodiments, the hexose has L-configuration. In some embodiments, the hexose anomeric configuration is alpha (α). In some embodiments, the hexose anomeric configuration is beta (β). In some embodiments, the hexose anomeric configuration is both alpha and beta.

[0077] With multiple hydroxyl groups on the ring, hexose increases the aqueous solubility of organic molecules. Because hexose is a natural product, its use as a prodrug form has advantages such as non-toxicity and high hydrophilicity. For example, glucuronide prodrugs as shown in Scheme 4:



Scheme 4. A glucuronide prodrug shown in **Scheme 4** comprises an N-pyridyl β -glucuronide moiety. β -Glucuronidase in the biological system is a lysosomal enzyme that plays a role in the degradation of glucuronic acid-containing glycosaminoglycans. It has a broad substrate specificity. Accordingly, β -glucuronidase recognizes the carbamate structure in the glucuronide prodrug and enzymatically cleaves the glycosidic bond to afford, the glucuronic acid, the freed amino-pyridine, and carbon dioxide as the enzymatic reaction products.

[0078] Similarly, a lysosomal enzyme such as β -galactosidase recognizes and cleaves galactoside β -prodrugs. Glucosidase is another example of enzymatic targets available for prodrug design.

Further Forms of Compounds Disclosed Herein

Isomers/Stereoisomers

[0079] In some embodiments, the compounds described herein exist as geometric isomers. In some embodiments, the compounds described herein possess one or more double bonds. The compounds presented herein include all cis, trans, syn, anti, entgegen (E), and zusammen (Z) isomers as well as the corresponding mixtures thereof. In some situations, the compounds described herein possess one or more chiral centers and each center exists in the R configuration, or S configuration. The compounds described herein include all diastereomeric, enantiomeric, and epimeric forms as well as the corresponding mixtures thereof. In additional embodiments of the compounds and methods provided herein, mixtures of enantiomers and/or diastereoisomers, resulting from a single preparative step, combination, or interconversion are useful for the applications described herein. In some embodiments, the compounds described herein are prepared as their individual stereoisomers by reacting a racemic mixture of the compound with an optically active resolving agent to form a pair of diastereoisomeric compounds, separating the diastereomers and recovering the optically pure enantiomers. In some embodiments, dissociable complexes are preferred. In some embodiments, the diastereomers have distinct physical properties (e.g., melting points, boiling points, solubilities, reactivity, etc.) and are separated by taking advantage of these dissimilarities. In some embodiments, the diastereomers are separated by chiral chromatography, or preferably, by separation/resolution techniques based upon differences in solubility. In some embodiments, the optically pure enantiomer is then recovered, along with the resolving agent.

Labeled compounds

[0080] In some embodiments, the compounds described herein exist in their isotopically-labeled forms. In some embodiments, the methods disclosed herein include methods of treating diseases by administering such isotopically-labeled compounds. In some embodiments, the methods disclosed herein include methods of treating diseases by administering such isotopically-labeled compounds as pharmaceutical compositions. Thus, in some embodiments, the compounds disclosed herein include isotopically-labeled compounds, which are identical to those recited herein, but for the fact that one or more atoms are replaced by an atom having an atomic mass or mass number different from the atomic mass or mass number usually found in nature. Examples of isotopes that can be incorporated into compounds of Formula (Ibb), or a solvate, or stereoisomer thereof, include isotopes of hydrogen, carbon, nitrogen, oxygen, phosphorous, sulfur, fluorine, and chloride, such as ^2H , ^3H , ^{13}C , ^{14}C , ^{15}N , ^{18}O , ^{17}O , ^{31}P , ^{32}P , ^{35}S , ^{18}F , and ^{36}Cl , respectively. Compounds described herein, and the metabolites, pharmaceutically acceptable salts, esters, prodrugs, solvate, hydrates or derivatives thereof which contain the aforementioned isotopes and/or other isotopes of other atoms are within the scope of this invention. Certain isotopically-labeled compounds, for example those into which radioactive isotopes such as ^3H and ^{14}C are incorporated, are useful in drug and/or substrate tissue distribution assays. Tritiated, i.e., ^3H and carbon-14, i.e., ^{14}C , isotopes are particularly preferred for their ease of preparation and detectability. Further, substitution with heavy isotopes such as deuterium, i.e., ^2H , produces certain therapeutic advantages resulting from greater metabolic stability, for example increased *in vivo* half-life or reduced dosage requirements. In some embodiments, the isotopically

swine, poultry, bovine, and equine animals.

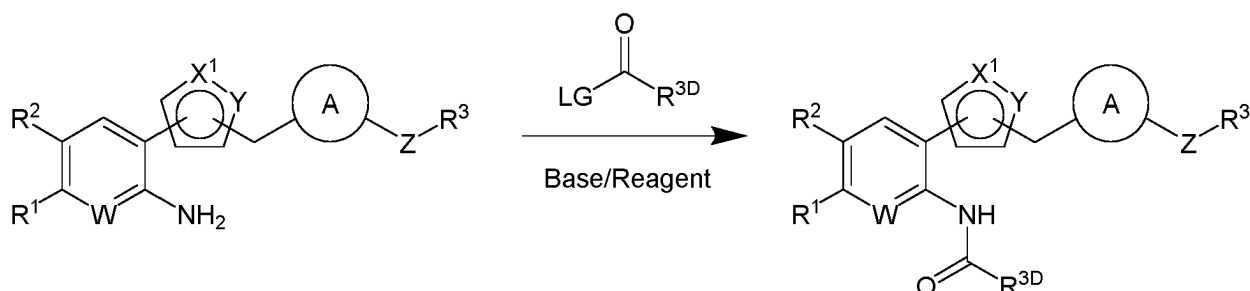
[0128] In some embodiments, are methods for treating a subject with a fungal disease comprising administering to the subject a combination treatment of a compound of Formula (Ibb), or a pharmaceutically acceptable salt, solvate, or stereoisomer thereof, and ketoconazole, wherein the subject is selected from the group consisting of cattle, sheep, goats, swine, poultry, bovine, and equine animals.

[0129] In some embodiments, are methods for treating a subject with a fungal disease comprising administering to the subject a combination treatment of a compound of Formula (Ibb), or a pharmaceutically acceptable salt, solvate, or stereoisomer thereof, and itraconazole, wherein the subject is selected from the group consisting of cattle, sheep, goats, swine, poultry, bovine, and equine animals.

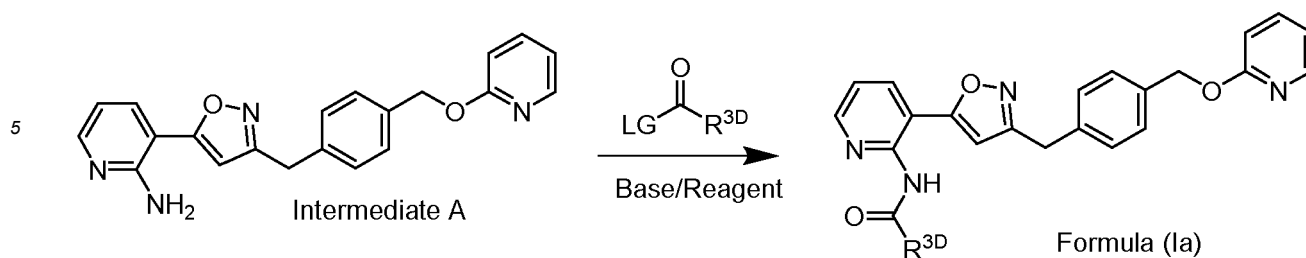
VI. Synthetic Methods

Method A

[0130]



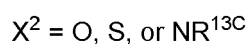
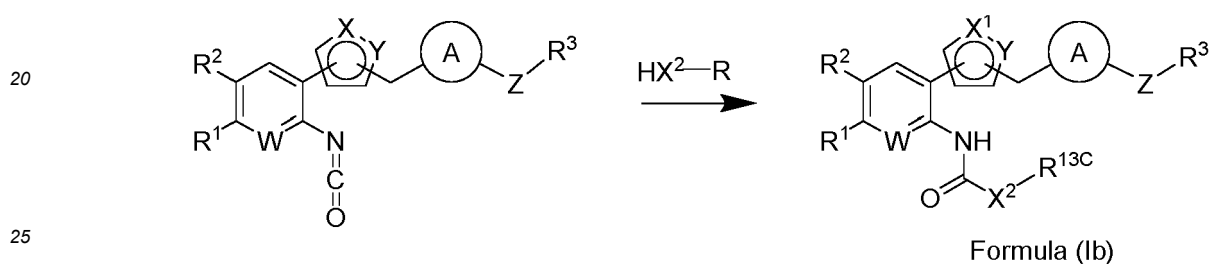
[0131] In Method A, one of X^1 and Y is nitrogen while the other is nitrogen or oxygen; W is N or N^+-R^{15} ; A is substituted or unsubstituted phenyl or substituted or unsubstituted pyridinyl; Z is a bond, $-(CH_2)_{z1}-$, $-O-$, $-S-$, $-CH_2O-$, $-OCH_2-$, $-NH-$, $-CH_2NH-$, $-NHCH_2-$, $-CH_2S-$, or $-SCH_2-$; $z1$ is 1 or 2; R^1 is hydrogen, halogen, $-NR^{1B}R^{1C}$, $-C(O)NR^{1B}R^{1C}$, $-OR^{1A}$, substituted or unsubstituted alkyl, or substituted or unsubstituted heteroalkyl; R^2 is hydrogen or halogen; R^3 is hydrogen, halogen, substituted or unsubstituted alkyl, substituted or unsubstituted heteroalkyl, substituted or unsubstituted cycloalkyl, substituted or unsubstituted heterocycloalkyl, substituted or unsubstituted aryl, or substituted or unsubstituted heteroaryl; R^{15} is $-SO_{n15}R^{15A}$, $-SO_{v15}NR^{15B}R^{15C}$, $-C(O)R^{15D}$, $-C(S)R^{15D}$, $-C(O)OR^{15D}$, $-C(S)OR^{15D}$, $-C(O)SR^{15D}$, $-C(O)NR^{15B}R^{15C}$, $-C(S)NR^{15B}R^{15C}$, $-CH_2OPO_3H_2$, $-CH_2OSO_3H$, $-CH_2OPO_3H$, $-CH_2OSO_3^-$, $-C(O)CH_2X^3$, or substituted or unsubstituted heteroalkyl; X^3 is $-F$, $-Cl$, $-Br$, $-I$, $-CHF_2$, $-CHCl_2$, $-CHBr_2$, $-CHI_2$, $-CF_3$, $-CCl_3$, $-CBr_3$, $-Cl_3$, or $-NO_2$; $n15$ is independently an integer from 0 to 4; $v15$ is independently 1 or 2; and R^{1A} , R^{1B} , R^{1C} , R^{3D} , R^{15A} , R^{15B} , R^{15C} , and R^{15D} are independently hydrogen, halogen, $-CF_3$, $-CCl_3$, $-CBr_3$, $-Cl_3$, $-COOH$, $-CONH_2$, substituted or unsubstituted alkyl, substituted or unsubstituted heteroalkyl, substituted or unsubstituted cycloalkyl, substituted or unsubstituted heterocycloalkyl, substituted or unsubstituted aryl, or substituted or unsubstituted heteroaryl; or R^{15B} and R^{15C} substituents bonded to the same nitrogen atom may optionally be joined to form a substituted or unsubstituted heterocycloalkyl or substituted or unsubstituted heteroaryl. As shown in Method A, an amine intermediate reacts with an acylating reagent in the presence of base and/or other reagent(s). The LG in the acylating reagent includes, but is not limited to, a leaving group, e.g., Cl and mixed anhydride, and OH . The reagent includes, but is not limited to, activation reagents, e.g., dicyclohexylcarbodiimide (DCC), (N-(3-dimethylaminopropyl)-N'-ethylcarbodiimide $\cdot$ HCl (EDAC $\cdot$ HCl), 1-hydroxybenzotriazole (HOBt), 4-(N,N-dimethylamino)pyridine (DMAP), (2-(1H-benzotriazol-1-yl)-N,N,N',N'-tetramethylaminium tetrafluoroborate/ hexafluorophosphate (TBTU/HBTU), (2-(7-aza-1H-benzotriazol-1-yl)-N,N,N',N'-tetramethylaminium tetrafluoroborate/ hexafluorophosphate (TATU/HATU), benzotriazol-1-yloxy-tris(dimethylamino)-phosphonium hexafluorophosphate (BOP), (benzotriazol-1-yloxy-tripyrrolidino-phosphonium hexafluorophosphate (PyBOP), and bromotripyrrolidino-phosphonium hexafluorophosphate (PyBrOP). The base includes, but is not limited to, pyridine, di-isopropylethylamine (DIPEA), trimethylamine (TEA), and N-methyl-morpholine (NMM). Method A 1



[0132] In some embodiments, as shown in Method A1, Intermediate A reacts with an acylating reagent in the presence of base and/or other reagent(s) to form a compound of Formula (Ia).

Method B

[0133]

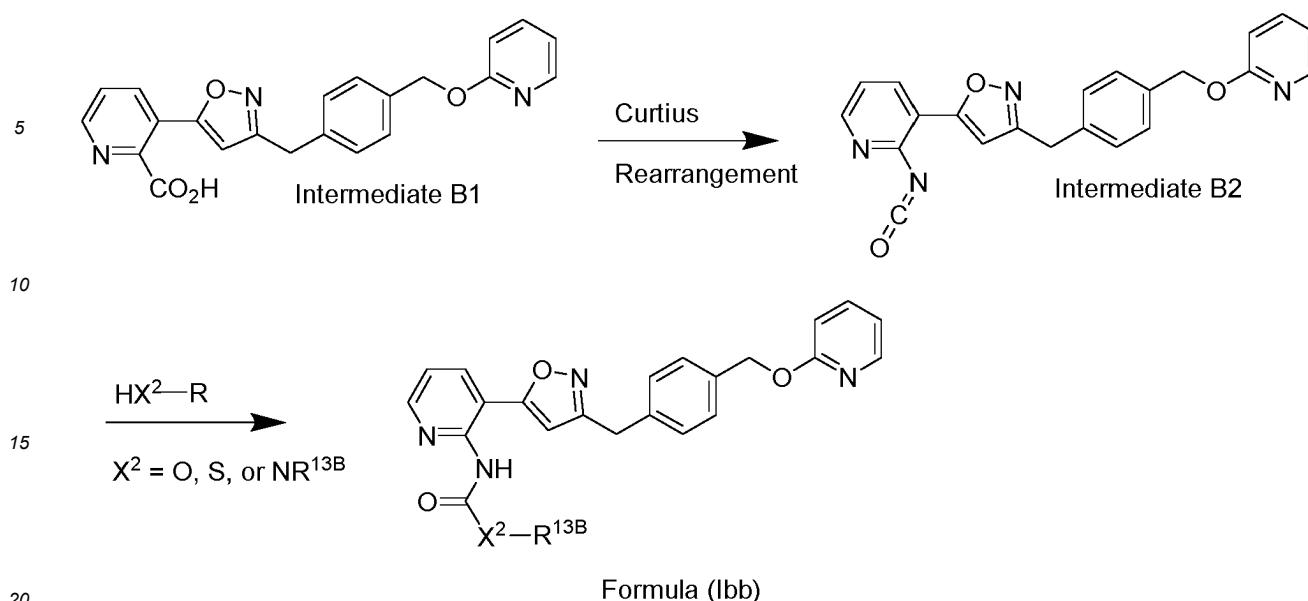


30 **[0134]** In Method B, R^1 , R^2 , R^3 , W , X^1 , Y , A and Z have the same meanings as in Method A; X^2 is $-O-$, $-S-$, or $-NR^{13B}$; and R^{13B} and R^{13C} are independently hydrogen, halogen, $-CF_3$, $-CCl_3$, $-CBr_3$, $-Cl_3$, $-COOH$, $-CONH_2$, substituted or unsubstituted alkyl, substituted or unsubstituted heteroalkyl, substituted or unsubstituted cycloalkyl, substituted or unsubstituted heterocycloalkyl, substituted or unsubstituted aryl, or substituted or unsubstituted heteroaryl; or R^{13B} and R^{13C} substituents bonded to the same nitrogen atom may optionally be joined to form a substituted or unsubstituted heterocycloalkyl or substituted or unsubstituted heteroaryl.

35 **[0135]** As shown in Method B, an isocyanate intermediate reacts with a nucleophilic reagent, e.g., an alcohol, thiol or free amine to afford a compound of Formula (Ib) that comprises, a carbamate, thiocarbamate, or urea moieties.

Method B1

[0136]

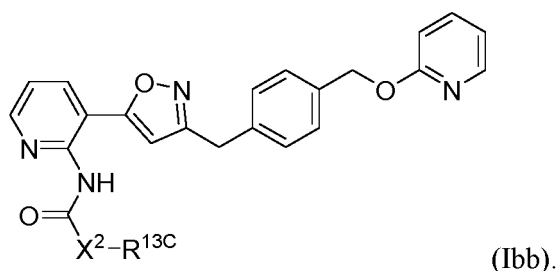


[0137] In some embodiments, as shown in Method B 1, Intermediate B 1, which bears a carboxylic group, undergoes a Curtius Rearrangement reaction to give Intermediate B2, which bears an isocyanate moiety. The Curtius Rearrangement conditions include, but are not limited to, (i) Intermediate B1 (1.0 eq.), Et_3N (1.3 eq.), $ClCO_2Et$ (1.5 eq.), $-10^\circ C$, THF, 2h; (ii) NaN_3 (1.7 eq.), $-10^\circ C$, H_2O , 1h; then toluene, reflux, 1 h. The Intermediate B2 bearing an isocyanate group reacts with an alcohol, thiol or free amine to afford a compound of Formula (Ibb). Conditions for reacting with the isocyanate group include, but are not limited to, nucleophilic reagent (HX^2-R^1 , 0.9 eq.), dioxane, reflux, 24 h.

Additional Embodiments

[0138] Embodiments include embodiments P'37 to P'41 following.

[0139] Embodiment P'37. The compound, or a pharmaceutically acceptable salt, solvate, tautomer, or stereoisomer thereof, wherein the compound has structural Formula (Ibb):



[0140] Embodiment P'38. A pharmaceutical composition comprising a pharmaceutically effective amount of the compound of Embodiment P'37, or a pharmaceutically acceptable salt, solvate, tautomer, or stereoisomer thereof, and at least one pharmaceutically acceptable carrier.

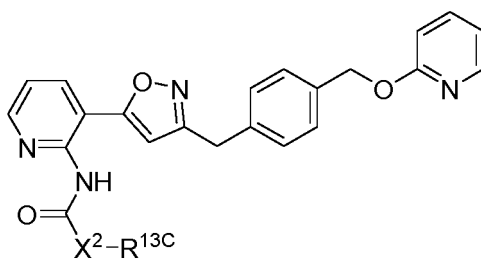
[0141] Embodiment P'39. A compound for use in a method of treating a fungal disease or infection, the method comprising administering to a subject in need thereof a pharmacologically effective amount of the compound of Embodiments P'37, or a pharmaceutically acceptable salt, solvate, tautomer, or stereoisomer thereof, or the pharmaceutical composition of Embodiment P'38.

[0142] Embodiment P'40. The compound for use in a method of Embodiment P'39, wherein the fungal disease or infection is a *Cryptococcus* disease or infection.

[0143] Embodiment P'41. The compound for use in a method of Embodiment P'39 or P'40, wherein the fungal disease or infection is azole-resistant and/or echinocandin-resistant.

[0144] Embodiments include embodiments 38 to 52 following.

[0145] Embodiment 38. The compound, or a pharmaceutically acceptable salt, solvate, tautomer, or stereoisomer thereof, wherein the compound has structural Formula (Ibb):



(Ibb).

[0146] Embodiment 39. A pharmaceutical composition comprising the compound of Embodiment 38, or a pharmaceutically acceptable salt, solvate, tautomer, or stereoisomer thereof, and at least one pharmaceutically acceptable carrier.

[0147] Embodiment 40. A compound for use in a method of treating a fungal disease or infection, the method comprising administering to a subject in need thereof a therapeutically effective amount of the compound of Embodiment 38, or a pharmaceutically acceptable salt, solvate, tautomer, or stereoisomer thereof, or the pharmaceutical composition of Embodiment 39.

[0148] Embodiment 41. The compound for use in a method of Embodiment 40, wherein the fungal disease or infection is caused by a *Cryptococcus*, *Aspergillus*, *Candida*, *Fusarium*, or *Scedosporium* fungus or a fungus from the Mucorales order.

[0149] Embodiment 42. The compound for use in a method of Embodiment 40 or 41, wherein the fungal disease or infection is azole-resistant and/or echinocandin-resistant.

[0150] Embodiment 43. The compound for use in a method of any one of Embodiments 40-42, further comprising administering at least one antifungal agent in combination with the compound of Embodiment 38, or a pharmaceutically acceptable salt, solvate, tautomer, or stereoisomer thereof, or the pharmaceutical composition of Embodiment 39.

[0151] Embodiment 44. The compound for use in a method of Embodiment 43, wherein the at least one antifungal agent is an azole, an echinocandin, amphotericin B deoxycholate, amphotericin B cochlolate, 5-fluorocytosine, terbinafine, griseofulvin, VL-2397, ibrexafungerp, orotomide F901318, or combinations thereof.

[0152] Embodiment 45. The compound for use in a method of Embodiment 44, wherein the azole is ketoconazole, fluconazole, posaconazole, itraconazole, voriconazole, isavuconazole, or miconazole.

[0153] Embodiment 46. The compound for use in a method of Embodiment 44, wherein the echinocandin is caspofungin, anidulafungin, micafungin, or rezafungin.

[0154] Embodiment 47. The compound for use in a method of any one of Embodiments 43-46, wherein the compound of Embodiment 38, or a pharmaceutically acceptable salt, solvate, tautomer, or stereoisomer thereof, or the pharmaceutical composition of Embodiment 39, and the antifungal agent are administered simultaneously, approximately simultaneously, or sequentially, in any order.

[0155] Embodiment 48. The compound for use in a method of Embodiment 47, wherein the compound of Embodiment 38, or a pharmaceutically acceptable salt, solvate, tautomer, or stereoisomer thereof, or the pharmaceutical composition of Embodiment 39, and the antifungal agent are administered simultaneously or approximately simultaneously.

[0156] Embodiment 49. The compound for use in a method of Embodiment 47, wherein the compound of Embodiment 38, or a pharmaceutically acceptable salt, solvate, tautomer, or stereoisomer thereof, or the pharmaceutical composition of Embodiment 39, or a pharmaceutically acceptable salt, solvate, tautomer, or stereoisomer thereof, and the antifungal agent are administered sequentially.

[0157] Embodiment 50. The compound for use in a method of Embodiment 49, wherein the compound of Embodiment 38, or a pharmaceutically acceptable salt, solvate, tautomer, or stereoisomer thereof, or the pharmaceutical composition of Embodiment 39, or a pharmaceutically acceptable salt, solvate, tautomer, or stereoisomer thereof, is administered before the at least one antifungal agent.

[0158] Embodiment 51. The compound for use in a method of Embodiment 49, wherein the compound of Embodiment 38, or a pharmaceutically acceptable salt, solvate, tautomer, or stereoisomer thereof, or the pharmaceutical composition of Embodiment 39, or a pharmaceutically acceptable salt, solvate, tautomer, or stereoisomer thereof, is administered after the at least one antifungal agent.

[0159] Embodiment 52. The compound for use in a method of any one of Embodiments 40-51, wherein the subject has reduced colony counts of fungi in the lungs after administration of the pharmaceutical composition.

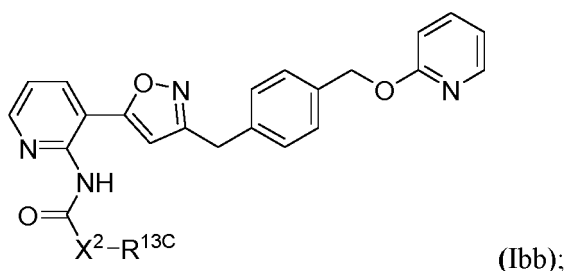
[0160] The disclosure will be further understood by the following non-limiting examples.

testinal, metabolic, urogenital, neurological, immunological, psychiatric, or neoplastic disorder with metastatic potential.

- History or presence of malignancy within the past year. Subjects who have been successfully treated with no recurrence of basal cell carcinoma of the skin or carcinoma in-situ of the cervix may be enrolled.
- Use of prescription medication within 14 days prior to the first dose of study drug and throughout the study.
- Use of non-prescription or over-the-counter medications within 7 days prior to the first dose of study drug and throughout the study.

Claims

- A compound of structural Formula (Ibb), or a pharmaceutically acceptable salt, solvate, tautomer, or stereoisomer thereof:



wherein:

X^2 is -O-, -S-, or -NR^{13B};

R^{13B} and R^{13C} are independently hydrogen, halogen, -CF₃, -CCl₃, -CBr₃, -Cl₃, -COOH, -CONH₂, substituted or unsubstituted alkyl, substituted or unsubstituted heteroalkyl, substituted or unsubstituted cycloalkyl, substituted or unsubstituted heterocycloalkyl, substituted or unsubstituted aryl, or substituted or unsubstituted heteroaryl; or R^{13B} and R^{13C} substituents bonded to the same nitrogen atom may optionally be joined to form a substituted or unsubstituted heterocycloalkyl or substituted or unsubstituted heteroaryl.

- The compound of claim 1, or a pharmaceutically acceptable salt, solvate, tautomer, or stereoisomer thereof, wherein X^2 is -O- and R^{13C} is substituted or unsubstituted phenyl.
- The compound of claim 1, or a pharmaceutically acceptable salt, solvate, tautomer, or stereoisomer thereof, wherein R^{13C} is hydrogen, substituted or unsubstituted alkyl, substituted or unsubstituted aryl, or substituted or unsubstituted heteroalkyl.
- A pharmaceutical composition comprising the compound of any one of claims 1-3, or a pharmaceutically acceptable salt, solvate, tautomer, or stereoisomer thereof, and at least one pharmaceutically acceptable carrier.
- A compound of any one of claims 1-3, or a pharmaceutically acceptable salt, solvate, tautomer, or stereoisomer thereof, for use in a method of treating a fungal disease or infection; optionally wherein:

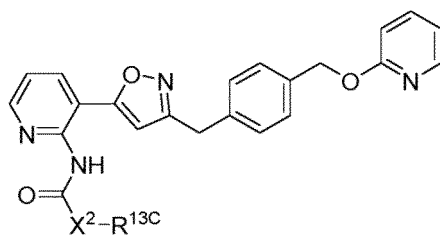
the fungal disease or infection is caused by a *Cryptococcus*, *Aspergillus*, *Candida*, *Fusarium*, or *Scedosporium* fungus or a fungus from the Mucorales order;

and/or the fungal disease or infection is azole-resistant and/or echinocandin-resistant;

wherein said treatment optionally further comprises administering at least one antifungal agent in combination with the compound of any one of claims 1-3.

Patentansprüche

- Verbindung der Strukturformel (Ibb) oder ein pharmazeutisch unbedenkliches Salz, Solvat, Tautomer oder Stereoisomer davon:



(Ibb);

wobei:

X^2 für -O-, -S- oder -NR^{13B} steht;

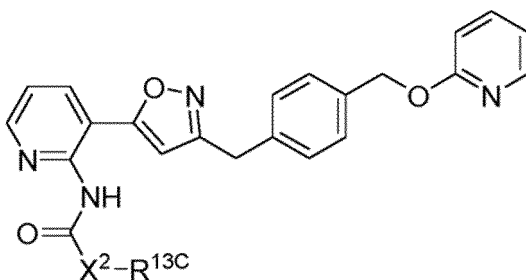
R^{13B} und R^{13C} unabhängig für Wasserstoff, Halogen, -CF₃, -CCl₃, -CBr₃, -Cl₃, -COOH, -CONH₂, substituiertes oder unsubstituiertes Alkyl, substituiertes oder unsubstituiertes Heteroalkyl, substituiertes oder unsubstituiertes Cycloalkyl, substituiertes oder unsubstituiertes Heterocycloalkyl, substituiertes oder unsubstituiertes Aryl oder substituiertes oder unsubstituiertes Heteroaryl stehen oder R^{13B}- und R^{13C}-Substituenten, die an dasselbe Stickstoffatom gebunden sind, gegebenenfalls zu einem substituierten oder unsubstituierten Heterocycloalkyl oder substituierten oder unsubstituierten Heteroaryl verbunden sein können.

2. Verbindung nach Anspruch 1 oder ein pharmazeutisch unbedenkliches Salz, Solvat, Tautomer oder Stereoisomer davon, wobei X^2 für -O- steht und R^{13C} für substituiertes oder unsubstituiertes Phenyl steht.
3. Verbindung nach Anspruch 1 oder ein pharmazeutisch unbedenkliches Salz, Solvat, Tautomer oder Stereoisomer davon, wobei R^{13C} für Wasserstoff, substituiertes oder unsubstituiertes Alkyl, substituiertes oder unsubstituiertes Aryl oder substituiertes oder unsubstituiertes Heteroalkyl steht.
4. Pharmazeutische Zusammensetzung, umfassend die Verbindung nach einem der Ansprüche 1-3 oder ein pharmazeutisch unbedenkliches Salz, Solvat, Tautomer oder Stereoisomer davon und mindestens einen pharmazeutisch unbedenklichen Träger.
5. Verbindung nach einem der Ansprüche 1-3 oder ein pharmazeutisch unbedenkliches Salz, Solvat, Tautomer oder Stereoisomer davon zur Verwendung bei einem Verfahren zur Behandlung einer Pilzerkrankung oder -Infektion; gegebenenfalls wobei:

die Pilzerkrankung oder -Infektion durch einen *Cryptococcus*-, *Aspergillus*-, *Candida*-, *Fusarium*- oder *Scedosporium*-Pilz oder einen Pilz der Ordnung Mucorales verursacht wird;
und/oder die Pilzerkrankung oder -Infektion Azolresistent und/oder Echinocandin-resistent ist;
wobei die Behandlung gegebenenfalls ferner das Verabreichen mindestens eines Antimykotikums in Kombination mit der Verbindung nach einem der Ansprüche 1-3 umfasst.

Revendications

1. Composé de formule structurale (Ibb), ou sel pharmaceutiquement acceptable, solvate, tautomère ou stéréoisomère de celui-ci :



(Ibb) ;

dans lequel :